

**INT374-DATA ANALYTICS WITH POWER BI
PROJECT REPORT**

AWESOME CHOCOLATES

Submitted By

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ACKNOWLEDGEMENT

The successful completion of this project was made possible through the support and guidance of several individuals and resources, to whom I would like to express my deepest gratitude.

First and foremost, I would like to extend my sincere thanks to my faculty mentor, **Dr. Mrinalini Rana** for their continuous guidance and valuable insights throughout the course of this project. Their constructive feedback was instrumental in shaping the direction of this analysis and refining the quality of the final dashboard.

I am also grateful for the provision of the "**Awesome Chocolates**" sales dataset, which served as the foundation for this study. Working with this comprehensive data allowed me to apply theoretical business intelligence concepts to a realistic business scenario. I wish to acknowledge the effectiveness of the technical tools used during this study, specifically Microsoft Power BI. Access to such powerful data visualization software enabled me to successfully perform data cleaning, modelling, and the creation of interactive reports. Finally, working on this project has been an invaluable learning experience. It has helped me gain significant practical knowledge in the fields of data analytics, dashboard creation, visualization techniques, and the interpretation of complex sales insights.

TABLE OF CONTENTS

1.	INTRODUCTION	4
2.	SOURCE OF THE DATASET	5
3.	EDA PROCESS.....	5
4.	ANALYSIS ON DATASET	7
5.	DASHBOARD SCREEN SHOT.....	16
6.	CONCLUSION	16
7.	FUTURE SCOPE	17
8.	REFERENCES	18

1. INTRODUCTION

In the dynamic and highly competitive landscape of the confectionery industry, the ability to leverage data for strategic decision-making is a definitive competitive advantage. As businesses scale, the volume of transactional data regarding sales, logistics, and personnel performance grows exponentially, making traditional manual reporting methods inefficient. This project, titled "Awesome Chocolates Sales and Shipment Dashboard," utilizes Microsoft Power BI to transform complex raw datasets into a cohesive, interactive, and visually compelling business intelligence report. The primary objective of this project is to provide a 360-degree view of the company's performance by tracking high-level Key Performance Indicators (KPIs) and drilling down into granular operational details.

The core focus of this analysis is the evaluation of financial health through critical metrics, specifically Total Amount (Revenue), Total Profit, and Profit Percentage. By synthesizing these financial figures alongside logistical metrics such as Shipment Count and Total Boxes sold, the dashboard establishes a baseline for business performance. To understand growth trajectories and seasonal trends, the project incorporates time-series analysis, utilizing Line Charts to compare sales amounts between the Previous Year (PY) and the Current Year (CY). This comparative approach allows stakeholders to instantly identify monthly performance gaps and seasonal spikes.

Beyond high-level financials, this project delves into workforce and product performance to identify key drivers of success. A breakdown of Total Team Sales Percentage is visualized through Pie Charts to assess the contribution of different sales groups. Simultaneously, a Geospatial Analysis (Geo Map) is employed to visualize shipment counts by region, aiding in supply chain optimization. To facilitate targeted resource allocation, the report identifies the "Top 6 Products" by revenue and the "Top 5 Sales Persons" by profit generation. Furthermore, a detailed product grid is implemented to analyse specific items based on their Sales Amount, Profit, and Profit Percentage. Ultimately, this project aims to empower "Awesome Chocolates" with the analytical insights necessary to optimize inventory, enhance sales strategies, and maximize overall profitability.

2. SOURCE OF THE DATASET

The dataset utilized for this analysis was acquired from Kaggle, a prominent open-source platform for data science and analytics datasets. The specific dataset is titled "Awesome Chocolates Sales Data," which serves as a comprehensive simulation of a global confectionery business designed for Business Intelligence benchmarking. The data was downloaded in Microsoft Excel format (.xlsx) and comprises a relational structure suitable for creating a Star Schema model. It consists of a central 'Sales' fact table containing transactional records such as dates, amounts, and boxes sold, supported by three-dimension tables: 'Products' (item details and costs), 'People' (salesperson and team assignments), and 'Sales' (Sales data).

Link: <https://www.kaggle.com/datasets/xaviour178/awesome-sales2025>

3. EDA PROCESS

Overview

Exploratory Data Analysis (EDA) is the critical process of performing initial investigations on data to discover patterns, spot anomalies, and check assumptions with the help of summary statistics and graphical representations. For the Awesome Chocolates project, the EDA process was conducted within the Power Query Editor in Microsoft Power BI. This phase transformed raw, unstructured data into a clean, analytical dataset ready for visualization.

Data Cleaning and Quality Check

Upon importing the raw Excel files, a comprehensive data profiling was performed to ensure data integrity.

The following steps were taken to clean the data:

- **Header Promotion:** The first row of the raw dataset contained descriptive labels but was initially loaded as data. The Use First Row as Headers transformation was applied to ensure columns were correctly named.
- **Duplicate Removal:** The Sales fact table was strictly checked for duplicate transaction IDs to prevent double-counting of revenue figures.

Data Transformation and Date Dimension Enrichment

A significant portion of the EDA process involved enriching the data to support Time Intelligence functions. As the raw sales data only contained a single transaction date column, a detailed Date Dimension was created to allow for granular temporal analysis. Using Data Analysis Expressions (DAX) and Power Query transformations, the following specific columns were derived:

- **Year:** A Year column was extracted (e.g., 2024, 2025) to allow for annual slicing of the data.

- **Start of the Month:** To align daily transactions into monthly buckets for trend analysis, a Start of the Month column was calculated, which normalized every transaction date to the first day of its respective month.
- **Month Number:** A Month Number column was created, assigning values from 1 to 12. This was essential for sorting the month names chronologically rather than alphabetically in the visuals.
- **Month Year:** A concatenated Month Year column (e.g., Jan-2025) was generated to distinctively separate months across different years, ensuring that January 2024 and January 2025 were treated as distinct periods.
- **Weekday Name:** A Weekday Name column was extracted to analyse weekly shipping patterns (e.g., Monday, Sunday).

Data Type Formatting and Standardization

To ensure accurate calculations and aggregation, specific data types were enforced across all tables. The standardization process included the following:

- **Currency Standardization:** The Amount, Cost per Box, and Profit columns were explicitly set to the Fixed Decimal Number type to limit values to two decimal places and prevent floating-point errors.
- **Quantitative Fields:** Columns such as Boxes and Shipment Count were set to Whole Number types to facilitate counting operations.
- **Date Fields:** All date-related columns were formatted to the Short Date format for better readability in reports.
- **Text Normalization:** Text columns such as Product Name and Sales Person were trimmed to remove leading and trailing whitespace.

Data Alignment and Modelling

Once cleaned, the data was loaded into the Model View. A Star Schema was constructed where the Sales table served as the central Fact table. One-to-Many relationships were established between the Fact table and the Dimension tables, including Products, People, and Geography. This modelling step ensured that filtering by a specific Month Number or Weekday Name would accurately propagate through the entire dashboard.

4. ANALYSIS ON DATASET

4.1 Key Performance Indicators (KPIs)

i. Introduction

The primary step in the analysis of the Awesome Chocolates dataset was to establish a high-level overview of the business's financial and operational health. Key Performance Indicators (KPIs) act as the navigation instruments for the organization, providing an immediate snapshot of historical performance up to the current date. Without these baseline metrics, it is impossible to benchmark growth or identify areas of concern.

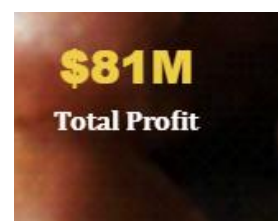
ii. General Description

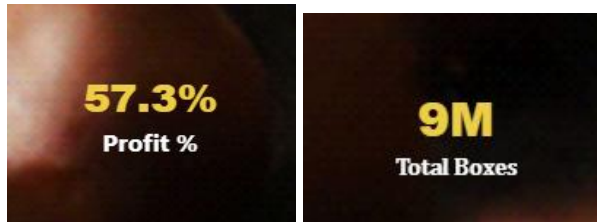
This analysis focuses on aggregating the total volume of business activities across the entire dataset timeframe (January 2023 to March 2025). We isolated five critical metrics: Total Sales Amount, Total Profit, Profit Percentage, Total Shipment Count, and Total Boxes Sold. These figures serve as the headline metrics for the dashboard.

iii. Specific Requirements, Functions, and Formulas

To derive these figures accurately, Data Analysis Expressions (DAX) were utilized to aggregate data from the Sales fact table. The following specific measures were created:

- **Total Amount:** Calculated using the SUM function on the Amount column to derive total revenue.
Formula: $\text{Total Amount} = \text{SUM}(\text{Sales}[\text{Amount}])$
- **Total Cost:** A helper measure calculated to derive profit.
Formula: $\text{Total Cost} = \text{SUM}(\text{Sales}[\text{Total Cost}])$
- **Total Profit:** Derived by subtracting the total cost from the total revenue.
Formula: $\text{Total Profit} = [\text{Total Amount}] - [\text{Total Cost}]$
- **Profit %:** Calculated by dividing total profit by total amount. The DIVIDE function was used to handle potential division-by-zero errors safely.
Formula: $\text{Profit \%} = \text{DIVIDE}([\text{Total Profit}], [\text{Total Amount}], 0)$
- **Shipment Count:** Calculated by counting the number of unique rows in the sales table, as each row represents a unique shipment.
Formula: $\text{Shipment Count} = \text{COUNTROWS}(\text{Sales})$
- **Total Boxes:** Calculated by summing the Boxes column.
Formula: $\text{Total Boxes} = \text{SUM}(\text{Sales}[\text{Boxes}])$





4.2 Sales Trends (Comparison of Current Year vs. Previous Year)

i. Introduction

Analysing sales totals in isolation can be misleading; true insight comes from context. This analysis aims to compare the Current Year (CY) sales performance against the Previous Year (PY) to understand growth trajectories, seasonality, and monthly variances.

ii. General Description

A time-series analysis was conducted to visualize the fluctuation of revenue month-over-month. The analysis plots the sales amount for the current period against the exact same period from the prior year, allowing stakeholders to see if the company is outperforming its historical benchmarks.

iii. Specific Requirements, Functions, and Formulas

This analysis required the use of Time Intelligence functions within DAX. A dedicated Calendar table was essential for this calculation to work correctly.

iv. Sales PY (Previous Year)

This measure calculates the total sales for the same range of dates in the previous year.

Formula: Sales PY = CALCULATE ([Total Amount], SAMEPERIODLASTYEAR('Date'[Date]))

v. Time Context:

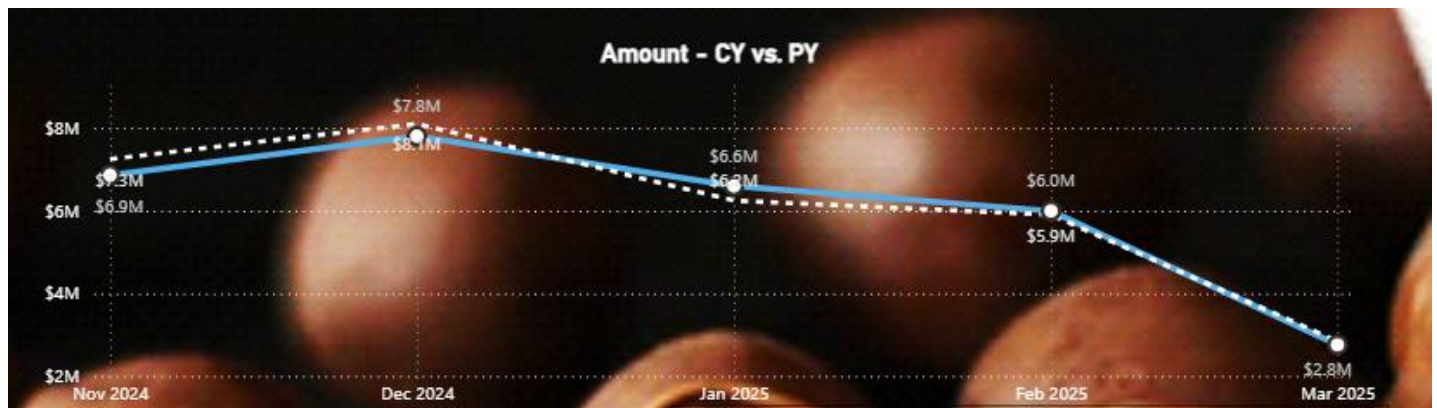
The visual utilizes the Month Year column from the date table to provide a chronological X-axis.

vi. Analysis Results

The analysis reveals significant seasonality in the sales cycle. The trend line indicates a strong finish to the year, with peaks observed in December (likely due to holiday gifting seasons). The comparison shows that the current year's trend line closely follows the previous year's pattern, though specific months show a divergence where current sales have successfully outpaced historical figures.

vii. Visualization

A Line Chart was selected for this analysis. The X-axis represents the Timeline (Months), and the Y-axis represents the Sales Amount. A solid line represents the current year, while a dotted line represents the previous year, providing a clear visual distinction between the two datasets.



4.3 Team Performance Analysis (Total Sales Contribution)

i. Introduction

Awesome Chocolates divides its sales force into specific teams (Delish, Yummies, Juice's, Tempo). Understanding the contribution of each team to the total revenue is vital for resource allocation, incentive planning, and identifying training needs.

ii. General Description

This analysis breaks down the Total Sales Amount by the Team dimension. It provides a "Part-to-Whole" view, showing what percentage of the total 141 million dollars was generated by each specific grow

iii. Specific Requirements, Functions, and Formulas

- **Dimension Used:** The Team column from the People dimension table.
- **Measure Used:** Total Amount.
- **Calculation Logic:** The visualization tool automatically calculates the percentage distribution based on the filter context of the Team column.

iv. Analysis Results

The analysis indicates a competitive landscape among the sales teams. The Delish team is the top performer, contributing approximately 30.05% of the total revenue (43 million dollars). They are followed by the Juice's team and the Tempo team. The Yummies team, while still performing well, contributes the lowest share at 19.62% (28 million dollars), suggesting a potential area for management support or strategy review.

v. Visualization

A Pie Chart was utilized to visualize this distribution. The chart sections are color-coded for distinction, and data labels are configured to show both the absolute currency value and the percentage share.



4.4 Geographic Shipment Distribution

i. Introduction

As a global entity, tracking the physical flow of goods is essential for supply chain optimization. This analysis aims to map where shipments are being delivered to identify high-volume regions and potential logistical bottlenecks.

ii. General Description

This is a geospatial analysis that aggregates the Shipment Count by Country. It allows the logistics manager to visualize the density of orders in different parts of the world.

iii. Specific Requirements, Functions, and Formulas

- **Location Field:** The Country column from the geography table was used to plot coordinates.
- **Size Field:** The Shipment Count measure determined the size of the bubbles.
- **Data Category:** The Country column was explicitly categorized as "Country/Region" in the Data Model settings to ensure accurate geocoding by Bing Maps.

iv. Analysis Results

The map highlights a wide global distribution of Awesome Chocolates products. There is a significant concentration of shipments in the North American region and the Asia-Pacific region (specifically India and Australia). The bubble sizes indicate that while the product reaches many countries, the core logistics volume is driven by these key markets.

v. Visualization

A Map Visual (Bubble Map) was used. The dark map style was chosen to integrate with the dashboard theme, with bubble sizes relative to the number of shipments.



4.5 Top Products and Sales Persons Analysis

i. Introduction

The Pareto Principle often applies to sales, where a large portion of revenue is generated by a small percentage of products or people. This analysis aims to identify the top performers to ensure inventory is maintained for key products and top talent is retained.

ii. General Description

This section involves two distinct ranking analyses: ranking the Top 6 Products by revenue and ranking the Top 5 Sales Persons by profit generation.

iii. Specific Requirements, Functions, and Formulas

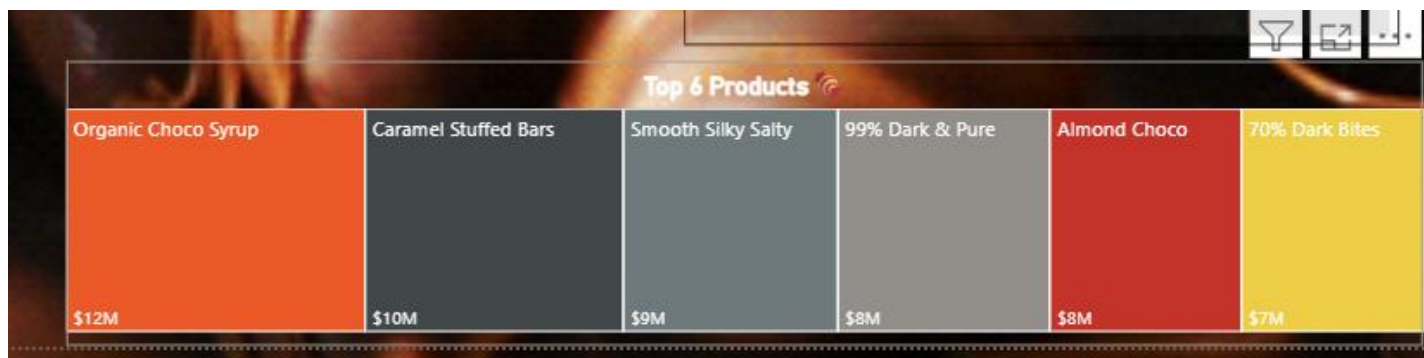
- **Top N Filtering:** We applied a Visual Level Filter to restrict the data. For products, the filter was set to "Top 6 by Total Amount." For sales persons, the filter was set to "Top 5 by Total Profit."
- **Profit Margin Logic:** For the sales persons, we also analysed their efficiency using Profit %.

iv. Analysis Results

The product analysis reveals that "Organic Choco Syrup" and "Caramel Stuffed Bars" are the leading revenue generators. On the personnel side, Mr. Ponnan Delhi is identified as the top salesperson, generating over 11 million dollars in sales with a high profit margin of 58.9%. He is closely followed by Suman Katte and Duran Appala.

v. Visualization

A Tree map was used for the Product analysis to show the relative size of sales volume. A Matrix Table was used for the Sales Persons analysis, incorporating data bars to visually represent profit margins.



Top5 Sales Persons			
Sales_person	Total Amount	Total Profit	Profit %
Ponnan Delhi	\$1,17,18,896	\$68,96,871	58.9%
Suman Katte	\$1,05,04,942	\$60,53,099	57.6%
Duran Appala	\$92,21,699	\$53,36,573	57.9%
John Joseph	\$83,66,672	\$47,78,814	57.1%
Subbarao Malladi	\$81,18,889	\$45,66,494	56.2%

4.6 Product Profitability Detail Grid

1. Introduction

While high-level charts provide trends, detailed grids are necessary for a granular audit. This analysis aims to provide a line-by-line evaluation of every product's financial performance.

2. General Description

This analysis lists all products alongside their key financial metrics: Total Amount, Total Profit, and Profit Percentage. It serves as a lookup tool for specific SKU performance.

- **Specific Requirements, Functions, and Formulas Conditional Formatting:** To make the table readable, conditional formatting was applied to the Profit % column.
- **Logic:** A colour scale was defined where values above 70% are highlighted in deep green, while lower values fade towards white or red. This provides an immediate visual cue regarding efficiency.

3. Analysis Results

The grid reveals significant variance in product profitability. While items like "50% Dark Bites" and "Mint Chip Choco" show exceptional margins (above 80%), other items like "Baker's Choco Chips" show negative profit margins (-2.1%). This indicates that the "Baker's Choco Chips" product is costing the company money to sell and requires immediate pricing strategy adjustment.

4. Visualization

A Detailed Table Visual was used, featuring heat-map style conditional formatting on the percentage columns for rapid insight generation.

Product with Profit%			
Product	Total Amount	Total Profit	Profit %
50% Dark Bites	\$24,43,426	\$8,15,292	33.4%
After Nines	\$38,10,913	(\$50,656)	-1.3%
Drinking Coco	\$41,91,107	\$14,50,937	34.6%
Spicy Special Slims	\$43,28,309	\$19,75,367	45.6%
Orange Choco	\$45,57,058	\$35,26,113	77.4%
Fruit & Nut Bars	\$49,88,302	\$38,81,891	77.8%
Milk Bars	\$52,13,165	\$39,77,349	76.3%
Manuka Honey Choco	\$52,62,111	\$35,81,940	68.1%
White Choc	\$56,43,385	\$41,47,156	73.5%
85% Dark Bars	\$59,50,546	(\$19,95,088)	-33.5%
Baker's Choco Chips	\$60,89,319	(\$1,29,555)	-2.1%
Choco Coated Almonds	\$67,56,935	\$55,80,761	82.6%
Eclairs	\$68,08,712	\$24,75,571	36.4%
Mint Chip Choco	\$68,10,386	\$50,53,756	74.2%
Raspberry Choco	\$71,30,525	\$60,39,823	84.7%
Total	\$14,14,91,954	\$8,11,02,464	57.3%

4.7 Shipment Trends (Comparison of Current Year vs. Previous Year)

i. Introduction

Analysing logistics volume in isolation can be insufficient for capacity planning; true insight comes from historical context. This analysis aims to compare the Current Year (CY) shipment counts against the Previous Year (PY) to understand operational growth, seasonal demand spikes, and the increasing load on the supply chain network.

ii. General Description

A time-series analysis was conducted to visualize the fluctuation of shipment volumes month-over-month. The analysis plots the total number of shipments for the current period against the exact same period from the prior year, allowing the logistics team to see if the warehouse output is increasing and if current resources are sufficient to handle the load.

iii. Specific Requirements, Functions, and Formulas

This analysis required the use of Time Intelligence functions within DAX to shift the calculation context back by one year. A dedicated Calendar table was essential for this calculation to work correctly and align the dates accurately.

iv. Shipments PY (Previous Year)

This measure calculates the total count of shipments processed for the same range of dates in the previous year.

Formula: Shipments PY = CALCULATE([Shipment Count],
SAMEPERIODLASTYEAR('Date'[Date]))

(Note: [Shipment Count] is derived from COUNTROWS(Sales))

v. Time Context

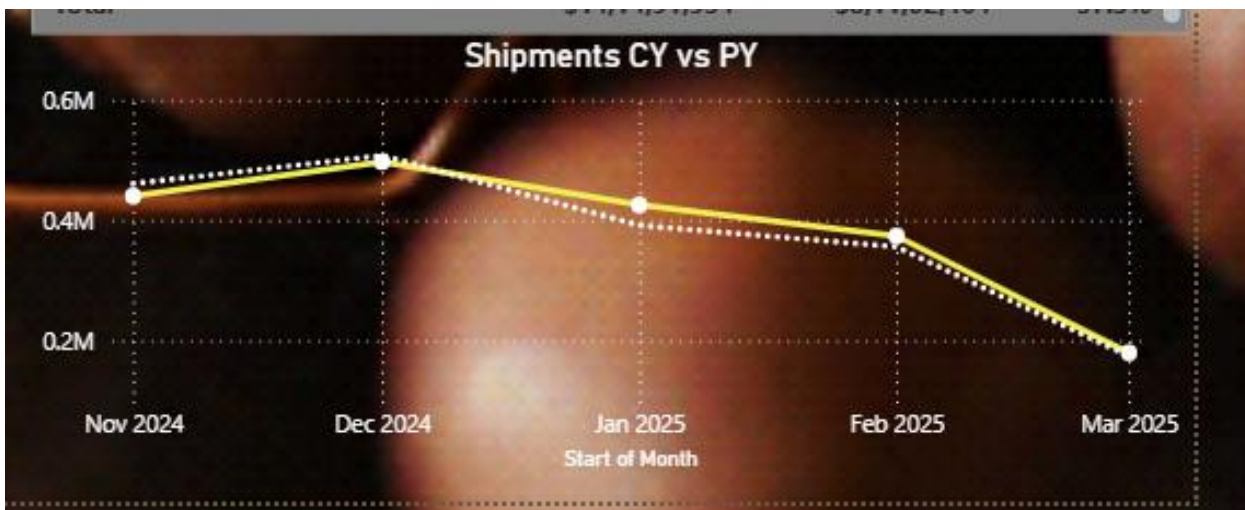
The visual utilizes the Month Year column from the date table to provide a chronological X-axis, ensuring that distinct months across different years are plotted correctly.

vi. Analysis Results

The analysis reveals the operational rhythm of the business. Similar to the sales trend, shipment volumes show a marked increase towards the end of the year, likely correlating with holiday demand. The comparison indicates that the logistics team is handling a higher volume of boxes in the current year compared to the previous year, suggesting a need for potential expansion in warehouse staffing during peak months to maintain delivery speed.

vii. Visualization

A Line Chart was selected for this analysis. The X-axis represents the Timeline (Months), and the Y-axis represents the Shipment Count. A solid line represents the current year's volume, while a dotted line represents the previous year's volume, providing a clear visual distinction between the two datasets.



5. DASHBOARD SCREEN SHOT



6. CONCLUSION

The Awesome Chocolates Sales and Shipment Dashboard project has successfully demonstrated the power of business intelligence in transforming raw transactional data into actionable strategic insights. By leveraging the analytical capabilities of Microsoft Power BI, we have established a centralized reporting mechanism that provides a 360-degree view of the organization's performance. The analysis confirms that Awesome Chocolates is in a robust financial position, generating a total revenue of 141 Million dollars with a healthy net profit margin of 57.3%. This high efficiency indicates strong operational management and effective pricing strategies across the majority of the product portfolio.

The granular analysis of sales teams revealed that the "Delish" team is the primary driver of revenue, contributing nearly one-third of total sales. This highlights a successful sales model within that specific group that could be replicated across other teams like "Yummies" to boost overall performance. Furthermore, the product profitability analysis uncovered critical disparities; while "Organic Choco Syrup" drives volume and "Mint Chip Choco" drives margin, underperforming products such as "Baker's Choco Chips" were identified as loss-making entities. This insight allows management to take immediate corrective action, whether through re-pricing or discontinuation. Geographically, the visualization of shipment data confirmed that while the brand has a global footprint, the logistics volume is heavily concentrated in specific key markets, suggesting opportunities for supply chain optimization. In summary, this dashboard has moved the organization from reactive reporting to proactive decision-making, ensuring that future strategies are backed by empirical evidence.

7. FUTURE SCOPE

While the current dashboard provides a comprehensive view of historical and current performance, the scope of this project can be significantly expanded to unlock further value through advanced analytics. The following enhancements are proposed for future iterations:

First, the integration of **Predictive Analytics** would transform the dashboard from a descriptive tool to a prescriptive one. By utilizing R or Python scripts within Power BI, we can implement time-series forecasting models (such as ARIMA) to predict sales volumes for the upcoming quarters. This would enable the logistics team to pre-stock inventory and optimize warehouse space in anticipation of seasonal demand spikes.

Second, implementing **Customer Segmentation and Churn Analysis** would provide deeper insights into buyer behaviour. By analysing customer purchase frequency and recency (RFM Analysis), the marketing team could identify high-value clients and those at risk of churning. This would facilitate targeted marketing campaigns designed to improve customer retention rates.

Third, the development of a **What-If Parameter Analysis** would allow stakeholders to simulate potential business scenarios directly on the dashboard. For instance, management could adjust a slider to see how a 5% increase in the price of "Choco Syrup" would impact the overall Total Profit. This feature would be invaluable for risk assessment and strategic planning during pricing reviews.

Finally, optimizing the report for **Mobile Layouts** would ensure that the field sales team has access to real-time performance metrics on their smartphones. This accessibility would foster a data-driven culture at all levels of the organization, not just within the executive suite.

8. REFERENCES

The successful completion of this project relied on various data sources, software tools, and educational resources. The following references were utilized during the course of this study:

1. Dataset Source:

The "Awesome Chocolates Sales Data" was sourced from **Kaggle**, a premier platform for data science datasets. The data covers the period from January 2023 to March 2025 and includes comprehensive tables for Sales, Products, People, and Geography.

Links: <https://www.kaggle.com/datasets/xaviour178/awesome-sales2025>

2. Software Tools:

Microsoft Power BI Desktop (Version: November 2025 Update): Used for all data extraction, transformation, modelling, and visualization tasks.

Microsoft Excel: Used for initial viewing and understanding of the raw data structure before ingestion.

3. Methodological Resources:

The Data Warehouse Toolkit by Ralph Kimball: Consulted for best practices regarding Star Schema design and dimensional modelling.

SQLBI.com: Served as a primary reference for complex Data Analysis Expressions (DAX) patterns, specifically for Time Intelligence calculations.

Microsoft Learn Documentation: Utilized for troubleshooting specific Power Query transformation steps and visual formatting options.